3月22日

微积分，共5题。

每一段旅途，每一段回忆，都是只属于旅人的、独一无二的奇迹。

祝顺利！

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1. $\lim\limits_{x\to 0}\left(\frac{1}{x^2}-\frac{1}{\sin^2 x}\right)=$

$W_1 = \lim\limits_{x\to 0}\frac{\sin^2 x - x^2}{x^4} = \lim\limits_{x\to 0}\frac{(\sin x + x)(\sin x - x)}{x^4} = \lim\limits_{x\to 0}\frac{2x\cdot(x-\frac{x^3}{6}-x)}{x^4} = -\frac{1}{3}$

$W_2 \overset{洛}{=} \lim\limits_{x\to 0}\frac{\sin 2x - 2x}{4x^3} \overset{洛}{=} \lim\limits_{x\to 0}\frac{2\cdot\cos 2x - 2}{12x^2} = \frac{1}{6}\lim\limits_{x\to 0}\frac{\cos 2x - 1}{x^2} = \frac{1}{6}\cdot(-2x^2) = -\frac{1}{3}$

$W_3 = \lim\limits_{x\to 0}\frac{\sin^2 x - x^2}{x^4} = \lim\limits_{x\to 0}\frac{2\xi(\sin x - x)}{x^4}$，$\xi$介于$(\sin x, x)$之间 $= \lim\limits_{x\to 0}\frac{2\xi\cdot(-\frac{1}{6}x^3)}{x^4} = -\frac{1}{3}$

2.设函数$f(x)$在点$x=3$的邻域内可微，且$\lim\limits_{x\to 3}f(x)=0$，$\lim\limits_{x\to 3}f'(x)=4016$。求$\lim_{x\to 3}\frac{\int_3^x\left[t\int_t^3 f(s)\,ds\right]dt}{(3-x)^3}$

$A = \lim\limits_{x\to 3}\frac{\int_3^x \varphi(t)\,dt}{(3-x)^3} \overset{洛}{=} \lim\limits_{x\to 3}\frac{x\int_x^3 f(s)\,ds}{-3(3-x)^2} = \lim\limits_{x\to 3}\frac{\int_3^x f(s)\,ds}{(3-x)^2} \overset{洛}{=} \lim\limits_{x\to 3}\frac{f(x)}{-2(3-x)}$

$= \lim\limits_{x\to 3}\frac{f(x)-f(3)}{x-3}\cdot\frac{1}{2} = \frac{1}{2}f'(3) = 2008.$

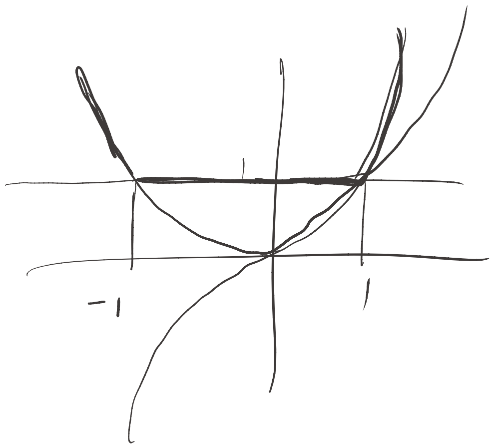
3.计算积分：(1) $\int\frac{dx}{1+\sin x+\cos x}$

记 $u=\tan\frac{x}{2}$. $\sin x = \frac{2u}{1+u^2}$，$\cos x = \frac{1-u^2}{1+u^2}$. $dx = \frac{2}{1+u^2}du$.

$I = \int\frac{1+u^2}{1+u^2+2u+1-u^2}\cdot\frac{2}{1+u^2}du = \int\frac{1}{1+u}du = \ln|1+u| + C = \ln\left|1+\arctan\frac{x}{2}\right| + C$

4.计算积分 $\int\max\{x^3,x^2,1\}\,dx$

$$f(x)=\begin{cases} x^2, & x<-1 \\ 1, & -1\le x\le 1 \\ x^3, & x>1 \end{cases}$$



$$\int f(x)\,dx = \begin{cases} \frac{1}{3}x^3 + C_1, & I_1 \end{cases}$$

$$\begin{cases} \quad\vdots \\ x + C_2, & I_2 \\ \frac{1}{4}x^4 + C_3, & I_3 \end{cases}$$

，为了保证可导，保证连续、

$$\begin{cases} -\frac{1}{3} + C_1 = -1 + C_2 \\ 1 + C_2 = \frac{1}{4} + C_3 \\ 1 + C_2 = -1 + C_2 \end{cases} \Rightarrow \begin{cases} C_1 = C - \frac{2}{3} \\ C_2 = C \\ C_3 = C + \frac{3}{4} \end{cases}$$

故原函数 $= \begin{cases} \frac{1}{3}x^3 + C - \frac{2}{3}, & I_1 \\ x + C, & I_2 \\ \frac{1}{4}x^4 + C + \frac{3}{4}, & I_3 \end{cases}$

5.已知$a < b$，且$a \cdot b > 0$，$f(x)$在$[a,b]$上连续，在(a,b)内可导。

试证：存在$\xi \in (a,b)$满足$\frac{1}{a-b}\begin{vmatrix} a & b \\ f(a) & f(b) \end{vmatrix} = f(\xi) - \xi \cdot f'(\xi)$

证：$\dfrac{af(b) - bf(a)}{b-a} = \dfrac{\frac{f(b)}{b} - \frac{f(a)}{a}}{\frac{1}{a} - \frac{1}{b}}$，令 $h(x) = \frac{f(x)}{x}$，$g(x) = \frac{1}{x} \neq 0$

由柯西中值定理得，

$$\frac{f(b)/b - f(a)/a}{\frac{1}{a} - \frac{1}{b}} = \frac{f'(\xi)\xi - f(\xi)/\xi^2}{1/\xi^2} = f'(\xi)\xi - f(\xi)$$，证毕.